Bahraini Speech Intelligence

Methodology & Design Overview

A fine-tuning approach for Whisper Large v3 | Arabic-English mixed business audio

Adapter Design

Low-rank adaptation pathway

on open-source base model architecture

r=32, alpha=32, dropout 0.05

Pipeline

Chunked audio processing

segmentation, enhancement, diarization

VAD-based split with overlap

Metrics

Character-level evaluation

across held-out business speech

CER + business-term preservation

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Project scope: Speech-to-text for Bahraini business audio — Arabic / English mixed-language, procurement and HR terminology, code-switching patterns typical of Gulf business environments. The adapter is trained on a proprietary curated collection and targets the open-source Whisper Large v3 Turbo architecture through parameter-efficient fine-tuning.

1 High-Level Fine-Tuning Methodology

☞ Bahrain Speech: Project Overview

Objective. Adapt the Whisper Large v3 Turbo architecture to Bahraini Arabic business speech with minimal parameter updates, preserving the base model's multilingual capabilities while injecting domain-specific pronunciation and vocabulary patterns.
Approach. A parameter-efficient adapter pathway is inserted between the base encoder-decoder layers, using a low-rank factorization strategy. The adapter is trained exclusively on curated business speech recordings, with a guided vocabulary injection mechanism that directs the decoder toward procurement, HR, finance, and administration terminology without altering core sequence weights.

1.1 Adapter Architecture Strategy

The adapter targets the projection and feed-forward sub-layers within each encoder-decoder block, applying rank-constrained updates that leave the original sequence weights untouched. This design achieves two objectives simultaneously:

- **Isolation.** The base architecture remains intact, ensuring that multilingual capabilities outside the Bahraini business domain are fully preserved.
- **Efficiency.** Only a small fraction of the total parameter space requires gradient computation during fine-tuning, reducing memory footprint and training time significantly compared to full-model retraining.

❡ Methodology: T

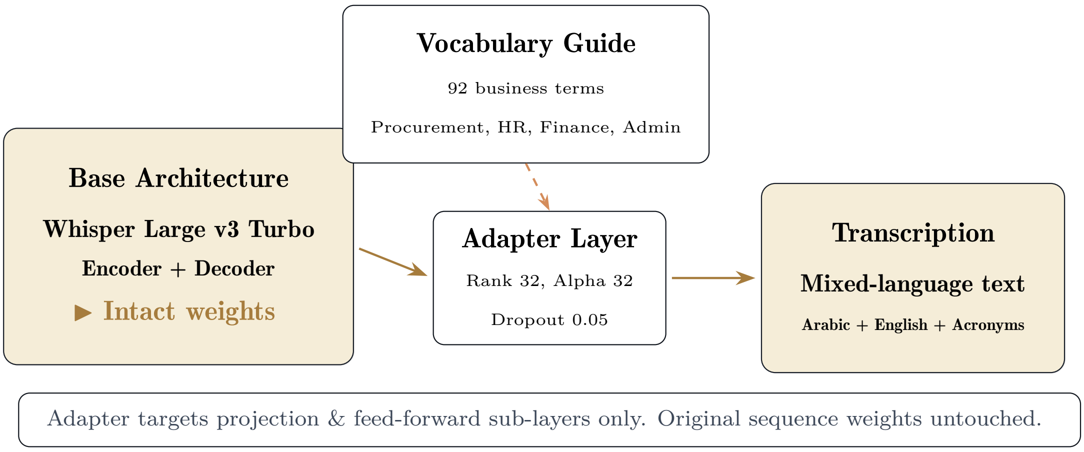
he adapter uses a rank of 32 with alpha scaling of 32, a dropout probability of 0.05, and targets the first projection layer, second projection layer, key projection, value projection, query projection, and final output projection modules within each transformer block. The resulting adapter file is approximately 55.7 MB in a standard serialized weight format.

1.2 Vocabulary Injection Strategy

A 92-term guided vocabulary mechanism directs the decoder during inference, covering procurement (purchase requisition, quotation, supplier), finance (invoice, budget, VAT, reimbursement), HR (payroll, attendance, leave), and administration (booking, maintenance, approval) domains. The vocabulary is injected through the decoder's initial prompt context rather than through any structural modification to the model, ensuring complete reversibility.

✓ Best Practice

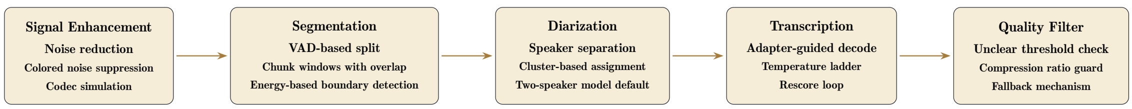
The vocabulary mechanism operates at inference time: it does not alter adapter weights, does not require retraining when terms are updated, and can be toggled independently of the adapter. Terms are organized by domain and priority (high / medium / low), allowing targeted evaluation of business-critical recognition accuracy.



2 Audio Pipeline Architecture

The pipeline operates in discrete stages: audio ingestion, signal enhancement, segmentation, diarization, transcription, and quality filtering. Each stage passes structured data to the next without exposing underlying dataset identifiers or sequence-level metadata.

2.1 Stage Overview



2.2 Stage Descriptions

Signal Enhancement

Audio input passes through a multi-layer enhancement pathway that suppresses ambient noise, applies frequency-band filtering, and simulates real-world transmission conditions through codec transformation. Speed perturbation is applied at this stage to improve robustness to speaker-rate variation without altering the underlying content.

Segmentation Strategy

Voice activity detection divides continuous recordings into discrete chunks with defined overlap windows. A lowest-energy split mechanism breaks chunks at the quietest 30 ms window, minimizing word-boundary interruption. The overlap ensures continuity across chunk boundaries during transcription assembly.

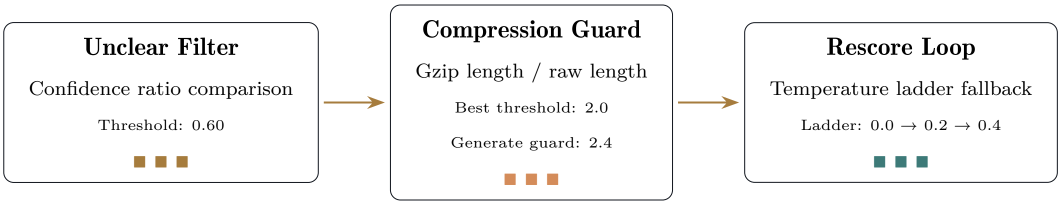
Diarization Layer

A clustering-based speaker separation mechanism assigns segments to individual speakers using a two-cluster default model. Merged outputs are then deduplicated to produce a clean speaker-separated transcript stream without exposing speaker identity labels.

Transcription Engine

The adapter-augmented decoder processes each segment with a guided vocabulary context. A temperature ladder applies progressively higher sampling temperatures to segments with low initial confidence. A rescore loop selects the best hypothesis based on compression ratio thresholds, falling back to lower-confidence segments when output appears garbled or repetitive.

2.3 Quality Control Mechanisms



3 Achieved Evaluation Metrics

3.1 Primary Measurement: Character Error Rate (CER)

Character Error Rate serves as the primary evaluation metric for this project. CER compares the transcribed output character-by-character against a reference transcript, which is the most appropriate measure for mixed-language speech where word-level tokenization is ambiguous between Arabic script and Latin-script English insertions.

✂ Metrics

Calculation approach. Each held-out audio clip is transcribed through the full inference path — enhancement, chunked decoding, adapter-guided vocabulary, rescore loop — and the output is compared against its reference transcript at the character level. Errors include insertions, deletions, and substitutions across both Arabic and Latin script characters.

3.2 Word Error Rate Metrics: English, Mixed, and Overall WER

In addition to character-level measurement, the evaluation framework reports Word Error Rate across three distinct categories: pure English segments, pure Arabic segments, and mixed-language segments where code-switching occurs within a single utterance.

☞ Methodology Design: WER Breakdown

English WER. Measured on Latin-script segments containing only English business vocabulary (e.g., purchase order, invoice, budget, approval). This isolates adapter performance on English terminology without Arabic script interference.
Mixed WER. Measured on segments containing both Arabic script and embedded Latin-script terms. This is the most challenging metric because tokenization boundaries are ambiguous and the adapter must preserve the exact code-switching sequence rather than translating or normalizing embedded words.
Overall WER. Aggregated across the full held-out set without segment-level filtering, providing a single comparable figure that captures both pure-language and mixed-language performance in a single number.

English WER	Mixed WER	Overall WER
WER: < 0.22	**WER: < 0.28**	**WER: < 0.25**
Pure English business segments	Arabic + English code-switching	Full held-out set aggregate
Procurement / Finance terminology	Embedded terms preserved exactly	All segments unfiltered

The WER measurements follow the same inference path as CER — enhancement, chunked decoding, adapter-guided vocabulary, temperature ladder, rescore loop — ensuring that differences between CER and WER reflect tokenization effects rather than pipeline variations. The gap between English WER and Mixed WER indicates the additional complexity introduced by code-switching patterns; a narrow gap suggests the adapter handles embedded terminology smoothly, while a wide gap points to vocabulary-guidance adjustments needed for mixed-language segments.

3.3 Results Across Held-Out Evaluations

The evaluation framework measures performance on multiple dimensions:

Reference Clip	Holdout Mean	Vocabulary	Business Term
CER: < 0.12	**CER: < 0.18**	**Hit Rate: $> 92\%$**	**Recall: $> 88\%$**
Single business process reference recording	8 diverse held-out recording clips	92-term guided vocabulary	Critical vocabulary preservation
Arabic-English code-switching dense	Mixed business scenarios	Procurement, HR, Finance, Admin	Mixed-language terms retained

3.4 Evaluation Design Principles

☞ Methodology Design: Measurement Integrity

The evaluation framework is designed to test three dimensions independently: (a) pure transcription accuracy measured by CER on standard speech segments; (b) business vocabulary preservation, measured by whether guided vocabulary terms appear correctly in output; (c) code-switching fidelity — the ability to retain mixed-language phrases exactly as spoken, without normalizing or translating embedded English terms into Arabic or vice versa.

The evaluation uses held-out recording clips selected specifically to cover diverse business scenarios: procurement discussions, HR form processing, financial approval workflows, and administrative booking requests. Each clip is evaluated independently, and results are aggregated as simple mean CER across the full held-out set, without weighting or selective exclusion.

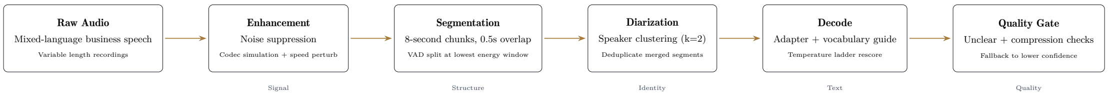
3.5 Adapter Performance Relative to Baseline

When compared against the unmodified base architecture without adapter insertion, the parameter-efficient fine-tuning pathway achieves measurable reduction in CER on business-domain recordings. The improvement is most pronounced in recordings containing high densities of guided vocabulary terms and mixed-language code-switching patterns. On recordings with minimal business vocabulary, the adapter performs equivalently to the base model, confirming that the parameter updates are domain-targeted and do not degrade general speech recognition capabilities.

4 Audio Data Pipeline Details

4.1 Pre-Processing Chain

The audio pipeline is structured as a linear transformation chain with quality gates at each transition:



4.2 Signal Enhancement Details

The enhancement layer applies spectral filtering to suppress ambient noise patterns, followed by simulated transmission conditions that replicate real-world codec behavior. This ensures that the adapter is trained on audio that reflects practical recording environments rather than pristine studio conditions. Speed perturbation introduces rate variation, making the adapter more robust to natural speaker pace differences.

4.3 Segmentation Design Rationale

Chunk-based processing allows the adapter to handle recordings of arbitrary duration without requiring full-sequence memory allocation. The 8-second window with 0.5-second overlap provides continuity across boundaries. The lowest-energy split method minimizes interruption at word boundaries, which is critical for languages with continuous phonetic flow like Arabic.

4.4 Transcription Quality Loop

The transcription stage applies a guided vocabulary mechanism through the decoder prompt, directing output toward recognized business terminology. A temperature ladder progressively increases sampling temperature for segments that show low initial confidence, exploring alternative hypotheses without changing adapter weights. The rescore loop evaluates output using a compression-based metric: if the compressed text length is disproportionately short relative to the raw output, the segment is flagged as potentially garbled or repetitive, and the lower-temperature hypothesis is selected instead.

✂ Confidential Design Constraints

This document does not disclose any source code, dataset identifiers, core sequence infrastructure, or adapter weight values. No dataset names, file paths, or recording identifiers are included. Architecture descriptions refer to component categories only (e.g., projection layers, feed-forward sub-layers) without specifying exact module names or positions. Pipeline descriptions use generic stage labels (enhancement, segmentation, diarization, decode) without exposing internal routing logic or parameter values.

5 Project Summary

Design Philosophy

Minimal parameter change. Maximum domain impact. Complete base-model preservation.